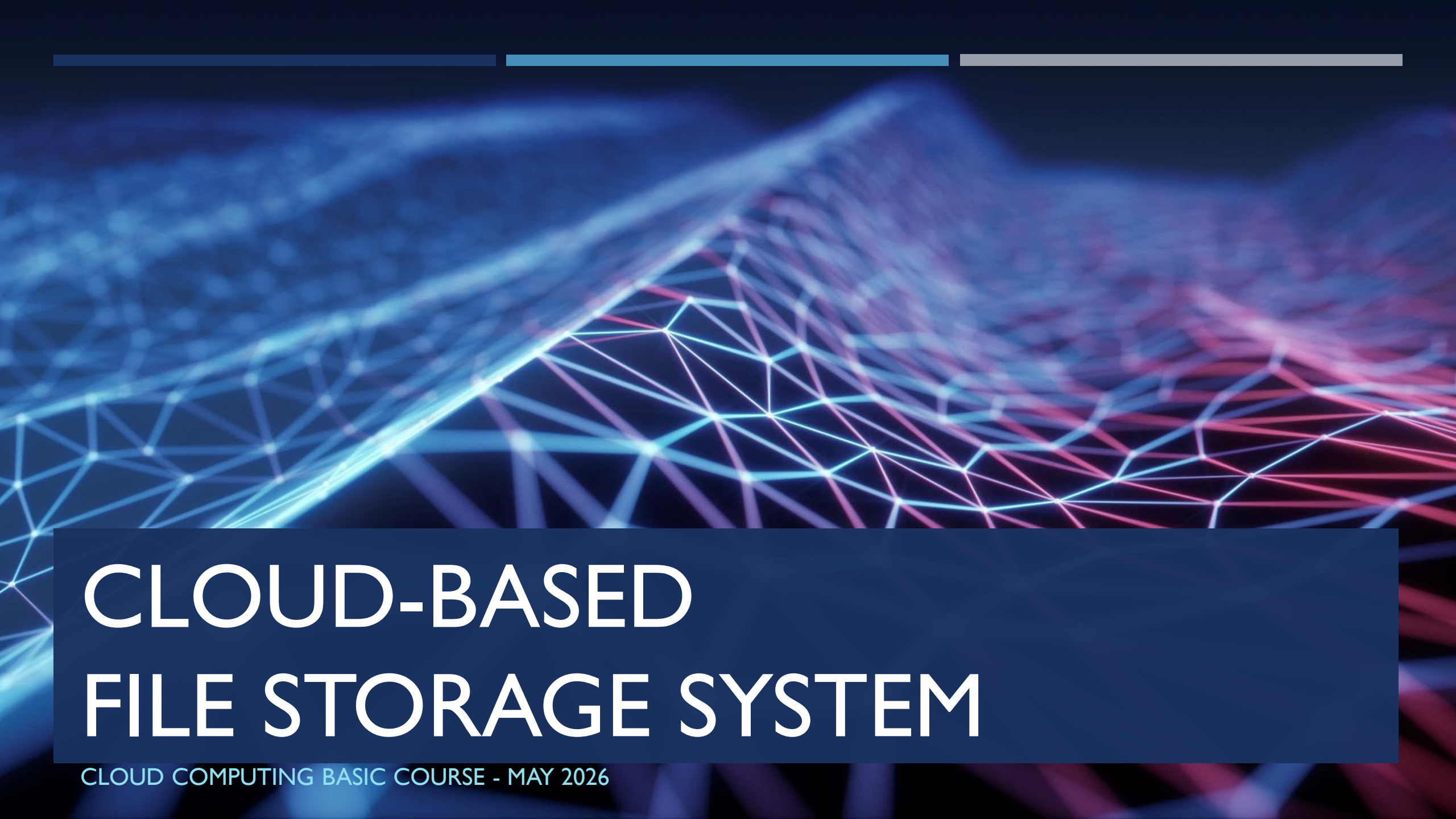




CLOUD-BASED FILE STORAGE SYSTEM

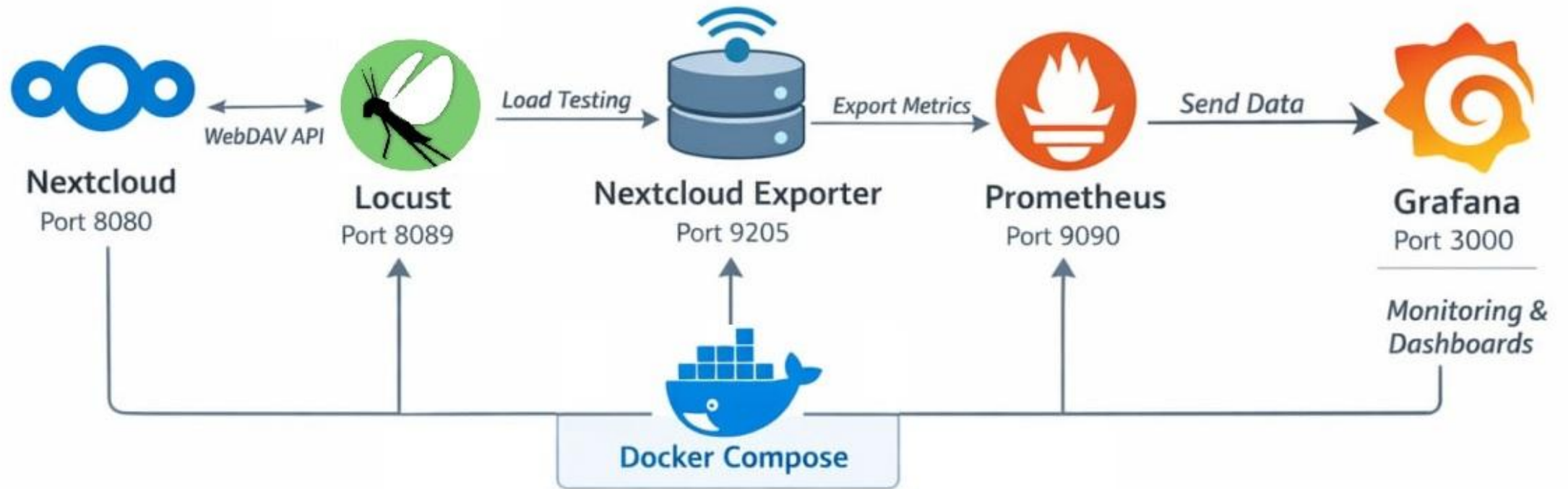
CLOUD COMPUTING BASIC COURSE - MAY 2026

PROJECT OVERVIEW

- **Objective:** Deploy a self-hosted cloud storage platform with monitoring and performance testing capabilities
- **Key Components:**
 - **Nextcloud:** File storage and user management
 - **Docker Compose:** Container orchestration
 - **Prometheus + Grafana:** Monitoring and visualization
 - **Locust:** Load testing and performance analysis
- **Requirements Addressed:**
 - ✓ User authentication & authorization
 - ✓ Private storage spaces
 - ✓ File operations (upload/download/delete)
 - ✓ Scalability, security, and cost-efficiency



SYSTEM ARCHITECTURE



NEXTCLOUD FEATURES



Accedi a Nextcloud

Accedi con nome utente o email

Password



→ Accedi

Hai dimenticato la password?

Accedi con un dispositivo

Admin Capabilities:

- ✓ User account creation and management
- ✓ Private storage allocation per user (`/remote.php/dav/files/username/`)
- ✓ Group management and permissions
- ✓ System configuration and security settings

User Capabilities:

- ✓ Upload/download/delete files
- ✓ File preview (images, PDFs, text)
- ✓ Folder creation and organization
- ✓ Web interface and desktop sync

To log in, hop on
<http://localhost:8080>

Nextcloud – un posto sicuro per tutti i tuoi dati

MONITORING STACK



Nextcloud Exporter

- Scrapes Nextcloud internal metrics
- Exposes data at ``http://localhost:9205/metrics``
- Key metrics: active users, file count, response times



Prometheus

- Time-series database
- Scrape interval: 15-30 seconds
- Configuration: ``prometheus/prometheus.yml``



Grafana

- Auto-provisioned Prometheus datasource
- Custom dashboards for real-time monitoring
- Access: ``http://localhost:3000``

LOAD TESTING WITH LOCUST

****Testing Framework: Locust****

****Test Configuration:****

- ****Script****: `locust-tasks/tasks.py`
- ****User simulation****: WebDavUser class
- ****Authentication****: Basic Auth (username/password)
- ****Wait time****: 1-3 seconds between tasks

****Test Tasks:****

```
``python
@task(2) # 40% weight
def upload(self): # PUT request
```

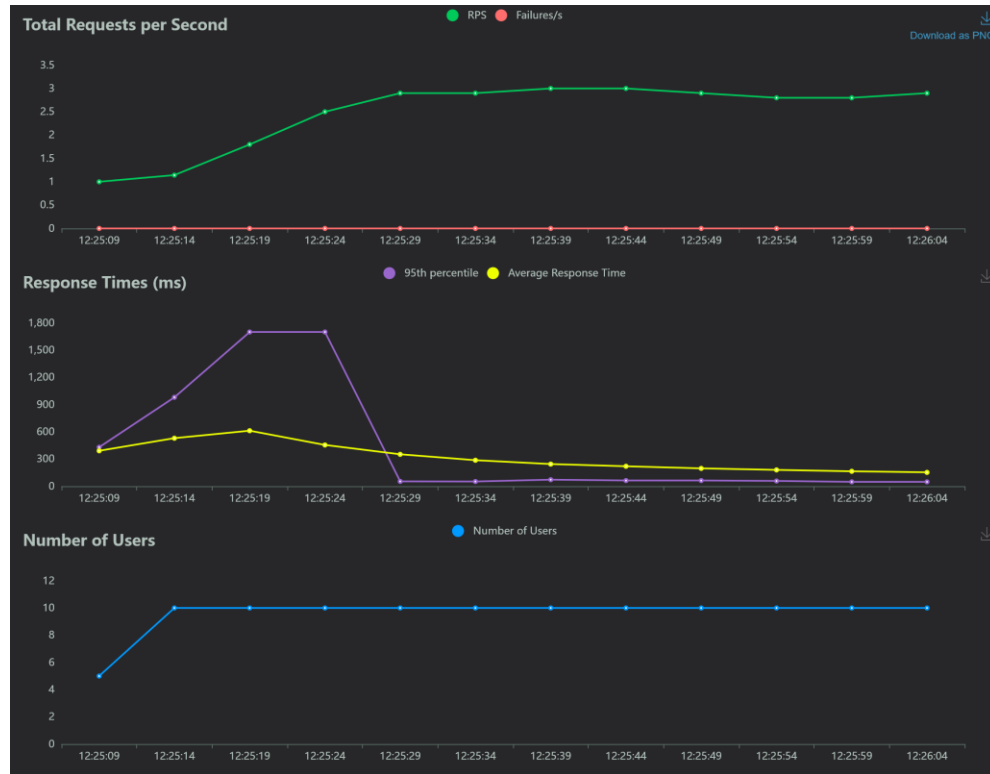
```
@task(2) # 40% weight
def download(self): # GET request
```

```
@task(1) # 20% weight
def delete(self): # DELETE request
``
```

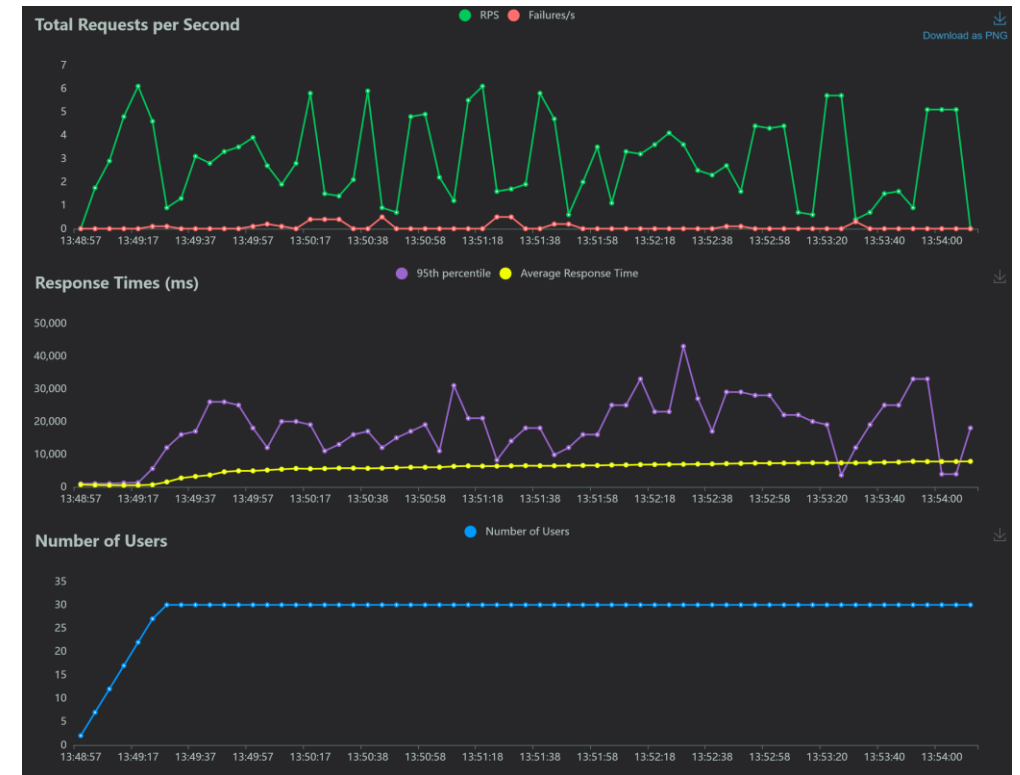
****Access****: `http://localhost:8089`

****[INSERT: Screenshot of Locust web interface]****

LOAD-TESTING RESULTS: 500KB - 5MB

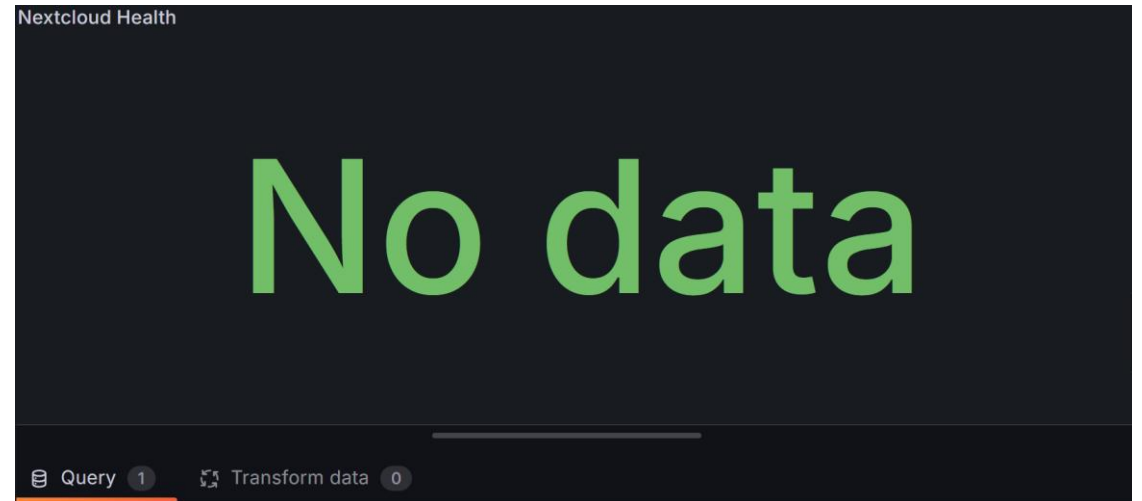


LOAD-TESTING RESULTS: 50MB



CHALLENGES & SOLUTIONS

- The **Nextcloud Exporter** was unable to pull metrics from Nextcloud.
 - Nextcloud was blocking the exporter because it was scraping too frequently.
 - This resulted in: `nextcloud_up = 0`
- WSL files from the **stress-test** almost **filled up the disk** with thousands of 50MB temporary files
 - deleting them didn't help!
 - The Windows diskpart tool saved the day
- *Many small issues... but I didn't give up!*



```
C:\Users>quit
"quit" non è riconosciuto come comando interno o esterno,
un programma eseguibile o un file batch.
```

ACHIEVEMENTS



✓ **Fully containerized Nextcloud deployment**



✓ **Complete monitoring stack** (Prometheus + Grafana)

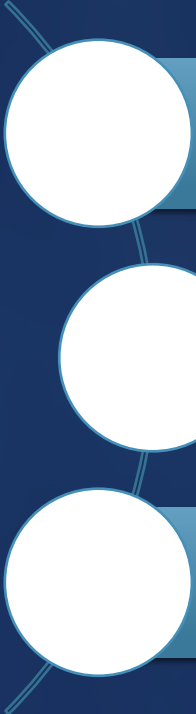


✓ **Automated load testing** with Locust



✓ **Comprehensive documentation** and **execution logs**

PERFORMANCE INSIGHTS



Small files (50KB) handled efficiently (~15ms response)

System **scales well** for moderate concurrent load

Authentication is critical for WebDAV operations

SCALING AND OTHER IMPROVEMENTS



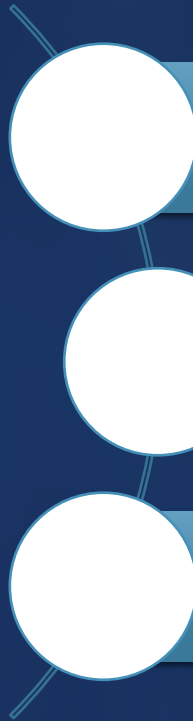
Scalability: PostgreSQL + Redis backend; horizontal scaling

Security: Add TLS/SSL via reverse proxy (Traefik/Nginx)

Cloud Deployment: AWS EC2 + S3 with auto-scaling

Larger Scale Tests: GB-sized files and 100+ concurrent users

COST-EFFICIENCY



All **open-source** components (zero licensing)

Lightweight containers optimize resource usage

Cloud deployment would use spot instances for cost optimization



THANK YOU



Component	Port	Purpose
Nextcloud	8080	Main application
Nextcloud Exporter	9205	Metrics exposure
Prometheus	9090	Time-series database
Locust	8089	Load testing
Grafana	3000	Visualization dashboard